

Open-Weight vs. Closed-Weight AI Models: Definitions, Licensing, Trade-offs, and Practical Guidance

Overview

The single most consequential design choice a modern AI company makes is not which architecture to use, but whether to publish the trained parameters of its model — the "weights" — for anyone to download, or to keep them locked inside a proprietary API. This choice, "open weight" versus "closed weight," has become the primary fault line in the AI industry, more consequential in practice than the older, stricter "open source" versus "proprietary" distinction inherited from software. By mid-2026, the gap between the two camps on raw capability has narrowed to a matter of months rather than years ([Stanford HAI AI Index Report 2026](#)), even as licensing terms, safety posture, and business models remain sharply divergent. This report defines the terms precisely, surveys the licensing spectrum with 2026 examples, compares performance and cost trade-offs, reviews the regulatory debate, and closes with practical guidance for builders choosing between the two.

What Model Weights Actually Are

A neural network is, at its core, a very large number of adjustable numerical parameters — "weights" — that determine the strength of connections between artificial neurons and shape how the model transforms an input into an output. During training, algorithms such as backpropagation repeatedly adjust these weights to minimize prediction error; the final, "locked-in" set of values is the trained model's entire learned "knowledge" ([Stanford HAI](#)). A large language model with, say, 70 billion parameters is a file containing roughly 70 billion such numbers (plus architecture metadata), typically stored at 16-, 8-, or 4-bit precision. Nothing about the training process, the curated dataset, or the code used to produce those numbers is contained in the weights file itself — it is purely the end-state artifact ([Open Source Initiative](#)).

This distinction matters because "open weights" refers narrowly to public release of that end-state parameter file, generally under a license that permits downloading, running, and often modifying and redistributing it. It says nothing about whether the training code, training data, or intermediate checkpoints are also released.

Open Weights vs. Open Source AI: The Critical Distinction

The terms are frequently used interchangeably in casual discussion and in marketing, but they describe materially different things, and the difference has become a point of genuine controversy in the AI industry.

Open weights means the final trained parameters are downloadable and usable, typically for inference and fine-tuning, but the training code, the full training dataset, and data-processing

pipelines are usually withheld. This makes the model impossible to fully audit, reproduce from scratch, or verify for embedded biases introduced during data curation ([Open Source Initiative](#)).

Open source AI, per the Open Source Initiative's formal Open Source AI Definition (OSAID) v1.0 finalized in October 2024, requires far more: the freedom to use the system for any purpose, to study how it works, to modify it, and to share it — applied not just to the model but to its weights, its code, and its data. OSAID specifically requires "sufficiently detailed information about the data used to train the system so that a skilled person can build a substantially equivalent system," including a full description of training data (or, if legally unshareable, a detailed accounting of why), a listing of all training data, and the complete training and inference source code, all under OSI-approved licenses ([OSI](#), [Open Source AI Definition 1.0](#)). Very few current frontier models meet this bar — most that are marketed as "open" are, by OSAID's standard, merely open-weight.

This gap is not academic. The OSI has explicitly and repeatedly stated that Meta's Llama models — despite Meta calling them "open source" — fail the definition on multiple counts: they discriminate against certain users (violating "freedom 0," the freedom to use for any purpose), restrict fields of use via an Acceptable Use Policy (violating point 6 on non-discrimination against fields of endeavor), and are non-transferable (creating problems under point 1 on free redistribution) ([OSI blog: "Meta's LLaMa license is still not Open Source"](#)). The Free Software Foundation reached a similar conclusion about the Llama 3.1 Community License Agreement. TechCrunch's coverage of the OSAID's finalization called it the field's first attempt to draw an "official" line under years of "open washing" — companies using "open source" as a marketing label for models that do not meet the traditional definition ([TechCrunch](#)).

Closed weight models, by contrast, never release the parameter file at all. Access is mediated entirely through a hosted API, a chat product, or a licensed enterprise deployment controlled by the vendor. The model runs exclusively on the provider's infrastructure (or infrastructure they explicitly authorize, such as a cloud partner), and users interact with it only through prompts and outputs — never touching the underlying weights.

The Licensing Spectrum

Real-world "openness" in AI sits on a continuum rather than a binary. From most to least permissive:

1. **Fully open source (OSAID-compliant or standard permissive licenses).** Weights, code, and (where legally possible) data are released under standard, OSI-approved licenses such as Apache 2.0 or MIT, with no usage restrictions beyond attribution. Most of Mistral AI's flagship releases fall here — the company states that models under Apache 2.0 can be used for any purpose, distributed freely, modified, and shared without restriction ([Mistral AI help center](#)). OpenAI's gpt-oss-120b and gpt-oss-20b, released in August 2025, are also under the "flexible Apache 2.0 license" ([OpenAI](#)). DeepSeek's newer flagships (V4-Pro, V4-Flash, V3.2, V3.1, R1) ship both code and weights under the MIT License, and Qwen3.8 was released with open weights under Apache 2.0 ([DeepSeek licensing guide](#); [The Decoder on Qwen 3.8](#)).
2. **Restricted/custom community licenses ("open-weight with strings attached").** The weights are downloadable and broadly usable, but a custom license imposes conditions: revenue

thresholds, user-count caps, geographic exclusions, output-usage restrictions, naming/attribution mandates, or revocable "acceptable use" policies. This is the category most associated with "open washing" controversy.

- **Meta's Llama** family uses a custom Community License requiring that any company (or its affiliates, aggregated across the entire corporate group, including >50%-owned subsidiaries) with more than 700 million monthly active users in the preceding calendar month must request — and may be refused — a separate commercial license from Meta before using Llama at all ([analysis of Llama 3.2 license terms](#); [Fortune](#)). The license also bars EU-domiciled individuals and companies from certain rights for Llama 3.2's multimodal models, requires "Built with Llama" branding, prohibits using Llama outputs to improve competing non-Llama models, is non-transferable (complicating M&A), and can be terminated by Meta for any breach with no cure period.
 - **Google's Gemma** family used a custom Gemma license (with Apache 2.0 introduced only for Gemma 4) that grants Google the right to "restrict (remotely or otherwise) usage" it deems violates its Prohibited Use Policy or applicable law, and requires any model trained even partly on Gemma outputs (a "Model Derivative") to inherit the same license terms ([Google Gemma Terms of Use](#); [TechCrunch on restrictive open licenses](#)).
 - **Mistral's non-production license (MNPL)**, introduced alongside Codestral, permits non-commercial and research use but requires a separate commercial agreement for production business use — a middle tier the company introduced explicitly to fund continued research while keeping other releases fully Apache 2.0 ([Mistral AI](#)). Separately, some Mistral models under a modified MIT license require companies with over \$20M in monthly revenue to obtain a commercial license or use Mistral's hosted platform ([Mistral help center](#)).
3. **Research-only / non-commercial licenses.** Weights are available to download and experiment with, but commercial deployment is explicitly prohibited or requires separate negotiation. Cohere has been explicit about supporting scientific but not commercial reuse of some of its releases ([TechCrunch](#)).
4. **Fully closed / proprietary, API-only.** No weights are ever released. The only access path is a paid API, subscription product, or specifically licensed enterprise deployment. OpenAI's GPT-5.x family, Anthropic's Claude Opus/Sonnet models, and Google's Gemini frontier models all sit here — served exclusively through hosted APIs and consumer/enterprise products ([OpenAI GPT-5](#); [Anthropic Claude models](#)).

Representative Examples as of 2026

Model family	Developer	Weight status	License type	Notes
Llama 4	Meta	Open weight	Custom Community License (700M-MAU carve-out, EU restrictions, non-transferable)	Marketed as "open source" but rejected as such by OSI (OSI)

Model family	Developer	Weight status	License type	Notes
Mistral (most models) / Mixtral	Mistral AI	Open weight	Apache 2.0 (some under modified MIT with \$20M revenue threshold, or MNPL for non-production)	Explicit advocate of openness as "safeguard for transparency" (Mistral AI)
Qwen 3 / 3.6 / 3.8	Alibaba	Open weight (mostly; some newer flagships reportedly closing)	Apache 2.0 for most releases	Alibaba's open models passed 3 billion downloads, surpassing Meta and Google by download count (Fortune/Bloomberg); some 2026 coverage suggests Alibaba is narrowing which flagship weights it releases openly
DeepSeek V4 / V3.2 / R1	DeepSeek	Open weight	MIT License (code and weights) for current flagships; older V3 base/Coder-V2/VL2 used a custom "DeepSeek Model License"	More permissive than Llama; DeepSeek runs a paid API/app atop the same downloadable models (DeepSeek licensing guide)
gpt-oss-120b / gpt-oss-20b	OpenAI	Open weight	Apache 2.0	OpenAI's first open-weight release since GPT-2; 120b runs on a single 80GB GPU, 20b on 16GB edge devices (OpenAI)
Gemma (1–3n) / Gemma 4	Google DeepMind	Open weight	Custom Gemma license (1–3n); Apache 2.0 for Gemma 4	Custom license drew criticism for revocable usage rights and derivative-model lock-in (TechCrunch)
Kimi K2 / K2.6 / K2.7	Moonshot AI	Open weight	Permissive open license (1T-parameter MoE, 32B active)	Strong agentic/coding benchmarks rivaling Claude and GPT-4.1 class models (Moonshot AI GitHub)
GLM-5 / GLM-5.1 / GLM-5.2	Zhipu AI (Z.ai)	Open weight	MIT License	1M-token context, ~750B total/40B active MoE, agent- and coding-first design (Eigent AI)
GPT-5 / GPT-5.4 / GPT-5.5	OpenAI	Closed weight	Proprietary, API/product only	Flagship reasoning model family, no downloadable weights (OpenAI)
Claude Opus/Sonnet 4.x	Anthropic	Closed weight	Proprietary, API/product only	Governed by Anthropic's Responsible Scaling Policy rather than public release (Anthropic RSP)
Gemini (frontier tiers)	Google DeepMind	Closed weight	Proprietary, API/product only	Google releases Gemma as the open sibling but keeps frontier Gemini closed; commentators have argued Google should open-weight a frontier Gemini model (Forbes)

Notably, Chinese labs (DeepSeek, Alibaba's Qwen, Moonshot AI, Zhipu/Z.ai) have emerged as the most prolific and permissive publishers of open-weight frontier-adjacent models, a strategic choice discussed further below.

Performance Gap: How Much Does Openness Cost You?

The performance gap between the best open-weight and best closed-weight models has narrowed dramatically. Stanford's 2026 AI Index reports that the top U.S. model's Arena Elo lead over China's largely open-weight competitors fell to just 2.7% by March 2026, down from a 17.5–31.6 point gap in May 2023 — a near-total convergence at the aggregate level ([Stanford HAI AI Index 2026, digested by Digital Applied](#); [Stanford HAI AI Index technical chapter](#)). Epoch AI's aggregate capability index similarly puts the open-weight frontier only about four months behind the closed frontier ([Gradient Ventures](#)).

However, the convergence is uneven and several important caveats apply:

- **Saturated public benchmarks mask a real gap on hard problems.** Both open and closed models now score above 90% on many named benchmarks, so headline "gap" numbers increasingly reflect benchmark saturation rather than genuine parity. The residual, meaningful gap shows up on the long tail: frontier reasoning (top-of-leaderboard math, competition coding, graduate-level science), calibrated safety/refusal behavior, resistance to jailbreaks, and coherence over very long multi-turn conversations — all roughly six to twelve months behind the closed frontier as of 2026 ([Nova AI Ops analysis](#)).
- **Private benchmarks tell a less flattering story than public ones.** Epoch AI's own analysis notes that open-weight models perform worse on private, non-gameable benchmarks than on public leaderboards — suggesting some degree of training to the test — and that leading labs do not always release their most capable models openly, meaning the "open frontier" measured by public leaderboards may not represent the true open-weight ceiling ([Gradient Ventures](#)).
- **Transparency is falling even as capability rises.** The Foundation Model Transparency Index average fell from 58 to 40 in the 2026 Index, as leading labs — across both open and closed camps — increasingly stopped disclosing training code, dataset sizes, and parameter counts, even for models whose weights are technically public ([Digital Applied digest of Stanford AI Index 2026](#)).
- **Where open models are competitive, they are very competitive.** OpenAI's own gpt-oss-120b achieves near-parity with OpenAI's proprietary o4-mini on core reasoning benchmarks while running on a single 80GB GPU, and gpt-oss-20b matches o3-mini-class performance on edge hardware with 16GB of memory — evidence that a lab can put a genuinely strong model into the open tier without necessarily matching its own top proprietary system ([OpenAI](#)). Kimi K2-Instruct's published benchmark table shows it beating DeepSeek-V3, Qwen3-235B, Claude Sonnet 4, Claude Opus 4, GPT-4.1, and Gemini 2.5 Flash on several coding benchmarks like LiveCodeBench and OJBench ([Moonshot AI GitHub](#)).

The practical read for builders: for the large majority of production tasks — chat assistants, RAG, summarization, extraction, standard coding — the open-weight frontier is close enough to the closed frontier that the choice is now driven mainly by cost, control, and compliance rather than raw capability. Only for the hardest reasoning, most safety-sensitive, or most adversarial use cases does the closed frontier retain a clear edge.

Trade-offs: Cost, Privacy, Customization, and Risk

Cost

Closed-weight APIs charge per-token, embedding vendor margin, R&D amortization, and safety infrastructure into the price. Open-weight models let an organization pay only for raw compute — but the crossover point where self-hosting beats API pricing arrives later than most teams expect, and depends heavily on utilization. One analysis benchmarking a self-hosted H100 against a budget open-weight API (DeepSeek V4 Flash at roughly \$0.14 per million input tokens) puts the breakeven at about 5.7 billion tokens per month; against a premium closed API (GPT-5-class, roughly \$5 per million tokens) the crossover arrives near 256 million tokens per month — but only assuming 60–70% sustained GPU utilization ([Digital Applied self-hosting guide](#)). Utilization is the hidden multiplier: at roughly 10% GPU utilization, effective cost-per-token can run about 10x the headline GPU rate, making an underused H100 more expensive per token than a premium proprietary API. Hidden costs beyond the GPU bill — DevOps salaries, model update cycles, infrastructure overhead — typically add a further 3–5x on top of raw hardware rental, according to the same analysis. Community discussion converges on a rough rule of thumb that self-hosting economics favor open weights once volume passes roughly 10 million to 100 million tokens per day, depending on workload ([Nova AI Ops](#); [Digital Applied](#)).

Privacy and On-Prem Deployment

This is where open weights have the clearest structural, durable advantage. Healthcare, finance, government, and other regulated sectors frequently cannot legally send data to a third-party API at all — open weights running on infrastructure the customer fully controls are, in many such cases, the only viable option, and this compliance gap is not expected to close as capability gaps do ([Nova AI Ops](#)). Fortanix frames the operational question builders eventually face: "Where is this model actually running? What happens to our data and prompts? And who can see the weights that power all of this?" — questions that closed, API-based models cannot answer to the satisfaction of many compliance and security teams, since prompts pass through a vendor's infrastructure whose logging, retention, and reuse-for-training practices the customer cannot fully verify or control ([Fortanix](#)). Even so, Fortanix notes that "closed weight" (in an enterprise sense, meaning vendor-controlled deployments) shifts rather than eliminates responsibility: teams choosing self-hosted open weights inherit full accountability for confidential inference, key management, and infrastructure hardening rather than outsourcing that risk to a vendor.

Fine-Tuning and Customization

Open weights can be fine-tuned, distilled, quantized, and modified without restriction (subject to license terms), enabling rapid domain adaptation, on-device deployment, and experimentation at the checkpoint level. Closed APIs increasingly offer managed fine-tuning services, but typically with restrictions on techniques, added latency, and no ability to inspect or export the resulting weights — the customization is real but bounded by what the vendor exposes ([Nova AI Ops](#)).

Safety and Misuse Risk

This is the most contested trade-off and the crux of the policy debate (detailed below). The core technical concern is that open weights, once released, cannot be recalled, and any built-in safety guardrails can be removed or bypassed through further fine-tuning or "jailbreaking," since the adversary controls the full model rather than only a filtered API surface ([Centre for Future Generations, "Beyond the Binary"](#)). Closed models let the developer maintain a persistent kill switch, monitor and rate-limit usage patterns, and patch or roll back the model in response to newly discovered risks — capabilities structurally unavailable once weights are public. OpenAI's own approach illustrates the industry's mitigation strategy for open releases: before shipping gpt-oss, OpenAI ran the models through comprehensive safety training and evaluations, then tested an adversarially fine-tuned version of gpt-oss-120b under its Preparedness Framework specifically to simulate a worst-case "already jailbroken" scenario, with results reviewed by external experts ([OpenAI](#)). Cisco's vulnerability analysis of open models similarly documents this as a live, empirically studied risk category rather than a hypothetical one ([Cisco AI blog, "Death by a Thousand Prompts"](#)).

The Regulatory and Policy Debate

The NTIA Report (United States)

Following President Biden's October 2023 Executive Order on AI, the NTIA (National Telecommunications and Information Administration) conducted a public consultation and produced the "Dual-Use Foundation Models with Widely Available Model Weights" report. Its central finding: open foundation models diversify the set of AI developers (including less-resourced actors), decentralize market control away from a handful of large labs, and let users avoid sharing sensitive data with third parties — but the same openness could produce harms to national security, equity, safety, privacy, or civil rights through misuse, oversight failures, or unclear accountability ([NTIA report](#)). Critically, the NTIA weighed three policy options — restricting the availability of model weights, continuously evaluating the ecosystem while building rapid-response capacity, or actively promoting openness — and recommended the middle path: **do not restrict the wide availability of model weights for dual-use foundation models at this time**, but actively monitor the ecosystem, maintain the capacity to respond quickly to specific risks, continue encouraging innovation and international coordination on open models, and preserve the explicit option to restrict certain classes of weights in the future if evidence warrants ([NTIA, Policy Approaches and Recommendations](#)).

The EU AI Act

The EU AI Act carves out a partial exemption for open-source/open-weight models: the Act generally does not apply to free and open-source models whose parameters are publicly available, except that even exempt providers must still comply with copyright law, publish a sufficiently detailed training-data summary, and meet the heightened obligations that apply to any model designated as posing "systemic risk" (a designation tied to factors including training compute and downstream reach), plus general AI value-chain responsibilities ([Open Future Foundation observatory](#)). The European Commission's guidelines for providers of general-purpose AI (GPAI) models, which entered into application on 2 August 2025, further clarify that only

those making "significant modifications" to a model — not those making minor changes — must comply with GPAI obligations, and specify the conditions under which open-source providers are exempted from certain requirements to preserve incentives for transparency and innovation ([European Commission digital strategy guidelines](#)). Enforcement powers for GPAI obligations, including fines, take effect from 2 August 2026, with providers whose models were already on the market before August 2025 required to reach compliance by 2 August 2027.

Safety Arguments on Both Sides

Proponents of continued open release argue: open weights let far more researchers, red-teamers, and smaller companies inspect, stress-test, and improve models than any single closed lab's internal safety team could; they reduce concentration of AI power in a handful of firms; and their marginal risk contribution — relative to what a well-resourced bad actor could already obtain from a closed model via jailbreaks or leaks, or build independently — may be smaller than the associated innovation and access benefits ([Stanford HAI issue brief, "Considerations for Governing Open Foundation Models," Bommasani, Kapoor, Klyman, Longpre, Ramaswami, Zhang, Ho, Schaaake, Narayanan, Liang](#)).

Critics counter that once weights are public, safety fine-tuning can be stripped out via further training, jailbreak resistance cannot be reliably guaranteed at scale, and the release is irreversible — there is no way to "recall" a model once millions of copies exist on external infrastructure ([Centre for Future Generations, "Beyond the Binary: A nuanced path for open-weight advanced AI"](#)). This 2026 report explicitly argues for moving past a simple open/closed binary toward a **tiered, safety-anchored release framework**, where the degree of openness for a given model is determined by rigorous, standardized risk assessment and demonstrated safety testing rather than by ideology or commercial incentive — and finds that current safety practices vary widely and inconsistently across developers and jurisdictions (EU, US, China, UK), with no commonly adopted release standard yet in place. A related *Nature* piece argues that releasing open-weight models in incremental steps, rather than all at once, could meaningfully reduce these risks while preserving most of the benefits ([Nature](#)). The UK AI Security Institute has published empirical work specifically measuring how far behind the closed frontier leading open-weight models lag on cyber-offense capability, treating this as a monitorable, quantifiable risk dimension rather than a purely theoretical one ([AISi](#)).

Anthropic's own posture illustrates the closed-weight safety rationale directly: its Responsible Scaling Policy ties increasingly capable model deployment to increasingly stringent safety and security requirements, calibrated to model capability tiers — a framework that is only enforceable because Anthropic retains full control over where and how its models run ([Anthropic, Responsible Scaling Policy](#)).

The Geopolitical Dimension

A distinct strand of the debate concerns why Chinese labs in particular have become the most prolific publishers of open-weight, frontier-adjacent models. Analysis from Takshashila argues China's open-weight strategy is driven by five reinforcing factors: lower training costs achieved through distillation and architectural efficiency; pursuit of geopolitical prestige and influence; a competitive push to undermine American labs' proprietary revenue models; the ability to

absorb heavy R&D costs through state-directed capital; and a long-term bid to drive global adoption of Chinese cloud infrastructure as the substrate on which these open models run ([Takshashila Institution](#)). The same analysis predicts this openness is not permanent, projecting that once domestic consolidation, ecosystem lock-in, and commoditization saturation thresholds are crossed — estimated around late 2028 — China may shift toward more graduated restrictions, including delayed international releases or gated access to its most capable frontier models. At the India AI Impact Summit and related APEC discussions in 2026, the U.S. and other nations reportedly signaled backing for open-source AI paired with "strong security" commitments, reflecting a broader international convergence toward treating openness and security as compatible rather than opposed goals ([CNBC](#); [Brookings on the India AI Impact Summit](#)). Alibaba's Qwen models alone reportedly passed 3 billion cumulative downloads by mid-2026, surpassing Meta and Google's open-model download counts ([Fortune/Bloomberg](#)).

Practical Guidance for Software Builders

Decision framework

- **Choose a closed-weight API (GPT-5.x, Claude, Gemini)** when: time-to-market matters most and infrastructure engineering capacity is limited; the task sits at the genuine capability frontier (hardest reasoning, safety-critical judgment, novel domains where open models still lag); volume is low-to-moderate (well under the token-volume breakeven for self-hosting); and there is no hard regulatory or contractual bar on sending data to a third-party processor.
- **Choose an open-weight model (self-hosted or via an open-weight-serving API provider)** when: data cannot legally or contractually leave your infrastructure (healthcare, finance, government, defense); token volume is high and sustained enough to clear the self-hosting cost breakeven (roughly tens of millions to low hundreds of millions of tokens per month, depending on model tier and target API price point); deep fine-tuning, distillation, or edge/on-device deployment is required; or vendor lock-in and long-term pricing risk are primary concerns.
- **Mixed/hybrid approaches** are common in 2026: using a closed frontier model for the hardest 5–10% of tasks (complex reasoning, agentic planning) while routing high-volume, lower-difficulty tasks to a self-hosted or open-weight-hosted model is an increasingly standard cost-optimization pattern, since the residual open/closed capability gap is concentrated in exactly the hardest tasks.

Licensing due diligence

Before adopting any "open" model commercially, actually read the license rather than trusting marketing language:

- Check for **usage-based revocation triggers** (Meta's 700M-MAU clause, Mistral's \$20M-monthly-revenue clause) that could force a later renegotiation as your product scales ([Llama license analysis](#); [Mistral licensing FAQ](#)).
- Check for **geographic exclusions** (Llama 3.2's EU carve-out for multimodal models).

- Check for **field-of-use restrictions** and acceptable-use policies that are incorporated by reference and can be unilaterally updated by the vendor (Gemma's remote-restriction right; Llama's Acceptable Use Policy).
- Check **transferability** if an acquisition or corporate restructuring is plausible — several custom licenses (Llama) are explicitly non-transferable.
- Prefer Apache 2.0 or MIT releases (Mistral's standard releases, gpt-oss, most DeepSeek V4-era models, Qwen3, GLM-5.2) when license simplicity and legal certainty matter, since custom licenses impose real legal-review overhead that deters many smaller companies from adoption at all ([TechCrunch](#)).

Hosting and inference options for open-weight models

- **Fully managed / serverless inference providers** (cloud marketplaces and dedicated inference platforms) let a team use an open-weight model's flexibility without operating GPUs directly — OpenAI's own gpt-oss models, for example, are available through AWS in addition to self-hosting ([AWS](#)).
- **Self-hosted serving stacks**: the 2026 default serving frameworks are **vLLM** (favored for high-throughput batch workloads) and **SGLang** (favored for prefix-heavy RAG and multi-turn agent workloads), after Hugging Face moved its earlier Text Generation Inference (TGI) framework to maintenance mode in December 2025 ([Digital Applied deployment guide](#)).
- **Confidential inference / secure enclaves**: for regulated industries running open weights on their own or a cloud vendor's infrastructure, confidential-computing approaches are increasingly treated as a requirement to fully realize the privacy benefits of open weights, since the underlying infrastructure still needs protection against insider or platform-level access ([Fortanix](#)).

Hardware requirements — sizing guide

Model memory requirements scale primarily with parameter count and quantization level, not with the license type. Representative figures for 2026:

Model class	Example	VRAM (Q4 quantization)	Typical hardware
Small (7–8B dense)	Mistral 7B, Llama-class small models	~5–6 GB	Single consumer GPU (RTX 4060/4070 class)
Edge MoE (~20B, low active params)	gpt-oss-20b	~16 GB	Single consumer GPU or high-end laptop/edge device (OpenAI)
Mid dense (70B)	Llama-class 70B	~35–40 GB (Q4)	Single 48–80GB datacenter GPU, or multi-GPU consumer setup
Large MoE (~120B total, ~5B active)	gpt-oss-120b	~64 GB (Q4_K_M); scales to 126–236 GB at Q8/FP16	Single 80GB GPU, or 128GB-unified-memory Mac/mini-PC (Hardwarepedia)
Frontier MoE (700B–1T total, tens of B active)	Kimi K2 (1T/32B active), GLM-5.2 (~753B/40B active)	Multi-hundred-GB range	Multi-GPU clusters or datacenter-scale serving; not practical on a single consumer machine (Moonshot AI ; Eigent AI on GLM-5.2)

Key sizing principles: **Mixture-of-Experts (MoE)** architectures like gpt-oss-120b, Kimi K2, and GLM-5.2 require memory capacity for the *entire* parameter count (all experts must be resident in memory) even though only a small fraction of parameters activate per token — meaning VRAM requirements track total parameters while inference *speed* tracks only active parameters, which is why large MoE models can feel fast once they fit in memory ([Hardwarepedia](#)). Context length also adds meaningfully to memory needs via the KV cache, which grows roughly linearly with context window size — sizing hardware for the actual context length needed (not just the base model weights) is essential for any RAG or long-document use case.

Comparison Summary Table

Dimension	Open-weight models	Closed-weight models
Weight access	Downloadable parameter files	Never released; API/product access only
Typical license	Apache 2.0, MIT, or custom community licenses with restrictions	Proprietary terms of service
Representative examples (2026)	Llama 4, Mistral/Mixtral, Qwen3.x, DeepSeek V4/R1, gpt-oss, Gemma, Kimi K2, GLM-5.2	GPT-5.x, Claude Opus/Sonnet 4.x, Gemini frontier tiers
Cost model	Pay for compute (self-host) or lower-cost hosted inference	Pay per token/subscription, vendor margin included
Data privacy / on-prem	Full control possible; only viable option for many regulated workloads	Data passes through vendor infrastructure; governed by vendor's data-handling policy
Fine-tuning / customization	Unrestricted (subject to license), full checkpoint-level control	Managed fine-tuning offered by some vendors, but bounded and non-exportable
Frontier capability (2026)	~Months behind closed frontier on public benchmarks; larger residual gap on hardest reasoning, safety behavior, long-horizon coherence	Leads on hardest reasoning tasks, calibrated safety, jailbreak resistance
Misuse / safety control	Cannot be recalled once released; safety training can be stripped via fine-tuning	Vendor retains kill switch, monitoring, and patch/rollback capability
Regulatory treatment	Partially exempted under EU AI Act (with conditions); US NTIA recommends monitoring, not restricting	Subject to full GPAI/systemic-risk obligations where applicable
Best fit	High-volume, compliance-bound, customization-heavy, cost-sensitive-at-scale use cases	Low-to-moderate volume, frontier-capability-dependent, fast-to-ship use cases

Conclusion

The open-weight/closed-weight split has evolved from a niche licensing debate into the central strategic axis of the AI industry. Precisely defined, "open weight" means only that the trained parameter file is downloadable — a narrower and more common condition than genuine "open source AI," which additionally requires open training code and data under the Open Source Initiative's formal 2024 definition, a bar that essentially no current frontier model, including Meta's Llama, actually clears. The practical licensing landscape spans a real spectrum from fully permissive Apache 2.0/MIT releases (Mistral, gpt-oss, current DeepSeek and Qwen models, GLM-5.2) through restricted community licenses laden with usage triggers and carve-outs

(Llama, Gemma) to fully closed, API-only proprietary systems (GPT-5.x, Claude, Gemini). By 2026, the raw capability gap between the best open and closed models has narrowed to a matter of months on most public benchmarks, even as it persists more durably on the hardest reasoning tasks and on safety behavior — while the structural advantages of open weights for cost-at-scale, data privacy, and customization remain undiminished and, in regulated industries, essentially permanent. Regulators in the US (NTIA) and EU (AI Act) have both settled, so far, on frameworks that preserve room for open release while reserving the right to intervene if evidence of serious harm accumulates, leaving the ultimate trajectory of the debate open and dependent on how the next generation of frontier-capable open-weight models is governed.